	$\frac{}{2}$
23	**Answer: A** $B = \frac{\mu_0 I}{2\pi r}$

	$F_{\text{onR}} = BI_RL = \frac{\mu_0 I}{2\pi r} I_R L$ For F due to P = F due to Q: $\frac{\mu_0 I_P}{2\pi r_P} I_R L = \frac{\mu_0 I_Q}{2\pi r_Q} I_R L$ Therefore $\frac{r_P}{r_Q} = \frac{I_p}{I_q}$ (does not depend on I_R nor direction of current) Since I_Q is twice I_P, $\frac{r_P}{r_Q} = \frac{1}{2}$ So position must be x to the left of P so that $\frac{r_P}{r_Q} = 0.5$
24	**Answer: D**